MACHINE LEARNING

Classification Algorithms

An Introduction for Students

KNN • SVM • Naïve Bayes • Decision Tree

What We Will Cover Today

01

K-Nearest Neighbors (KNN)

Classify by distance to neighbors

03

Naïve Bayes

Probabilistic classification using Bayes' theorem

02

Support Vector Machine (SVM)

Find the optimal separating hyperplane

04

Decision Tree

Rule-based tree structure for decisions

All four are supervised classification algorithms

01
K-Nearest Neighbors

KNN

"Tell me who your neighbors are, and I'll tell you who you are."

K-Nearest Neighbors (KNN)

KNN

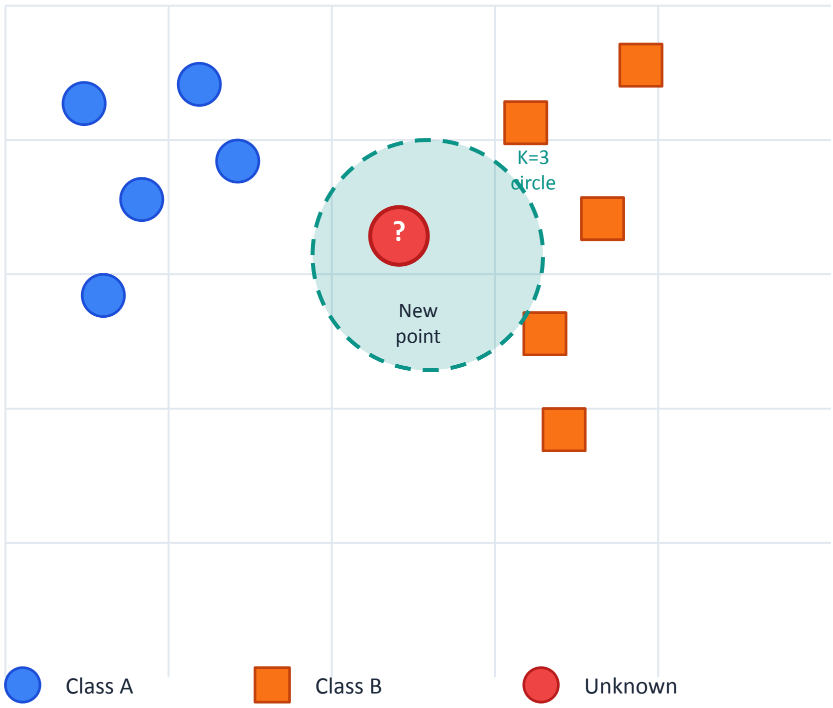
Core Concept

- Stores all training data (lazy learner)
- For a new point, find the K closest training examples
- Majority vote among neighbors determines the class
- Distance measured using Euclidean, Manhattan, or Minkowski

Key Parameters

K = number of neighbors | Small K ⟶ overfitting | Large K ⟶ underfitting

KNN – Visual Example



02

Support Vector Machine

SVM

"Find the widest street between two neighborhoods."

Support Vector Machine (SVM)

SVM

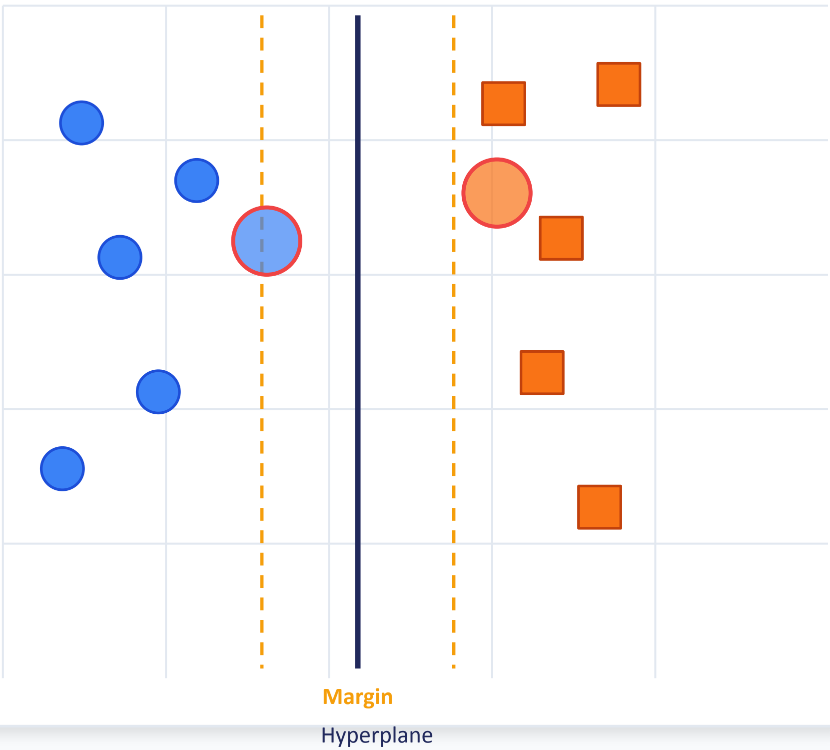
Core Concept

- Finds the optimal hyperplane that separates classes
- Maximizes the margin between classes
- Support vectors are data points closest to the hyperplane
- Uses kernel trick for non-linearly separable data (RBF, Polynomial)

Kernel Types

Linear | **RBF** | Polynomial | Sigmoid

SVM – Hyperplane & Margin



03
Naïve Bayes

$$P(A|B) = P(B|A) \times P(A) / P(B)$$

"Use what you already know to predict what you don't."

Naïve Bayes Classifier

BAYES

Core Concept

- Based on Bayes' Theorem of conditional probability
- Assumes features are independent (the 'naïve' assumption)
- Calculates probability of each class given the input features
- Very fast and works well with text/spam classification

Bayes' Theorem

```
P(Class | Features) = P(Features | Class) × P(Class) / P(Features)
```

Spam Filter Example

Word	P(word|Spam)	P(word|Ham)
Free	0.90	0.05
Money	0.85	0.10
Meeting	0.10	0.75
Project	0.15	0.80

→ Classify email as Spam if P(Spam|words) > P(Ham|words)

04

Decision Tree

Root ⟶ Branches ⟶ Leaves

"Ask the right questions in the right order."

Decision Tree

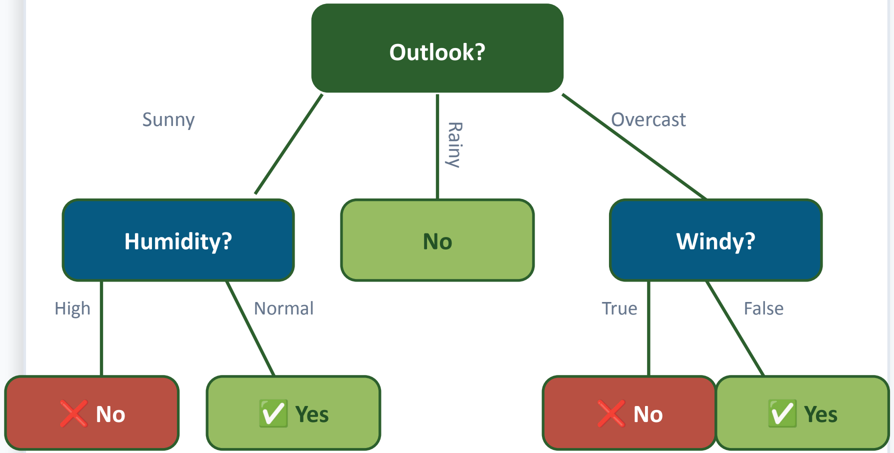
TREE

Core Concept

- Tree-shaped model of decisions and outcomes
- Each internal node represents a feature/attribute test
- Each branch represents the outcome of the test
- Leaf nodes represent class labels
- Split criterion: Information Gain or Gini Impurity

Advantage: Easy to understand & interpret (white-box model)

Play Tennis?



Algorithm Comparison

Algorithm	Type	Training Speed	Prediction Speed	Interpretable?	Best For
KNN	Lazy	⚡ Fast	🐢 Slow	Moderate	Small datasets, recommenders
SVM	Eager	🐢 Slow	⚡ Fast	Low	High-dim data, text, images
Naïve Bayes	Probabilistic	⚡ Fast	⚡ Fast	High	Text classification, spam
Decision Tree	Rule-based	⚡ Fast	⚡ Fast	Very High	Tabular data, explainability

KEY TAKEAWAYS

Summary

KNN	Memory-based, simple, but slow at prediction time
SVM	Powerful for complex boundaries, great with kernels
Naïve Bayes	Probabilistic, fast, ideal for text & NLP tasks
Decision Tree	Interpretable, easy to visualize & explain

No single algorithm wins — choose based on data size, interpretability needs, and task type.